

**JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION
JA-00732 – SPRDL1-24-0009**

1. Contracting Agency: Defense Logistics Agency - DLA Land Warren (DLA-WRN).

2. Description of Action: DLA-WRN proposes to award a New Firm-Fixed-Price contract to Oshkosh Defense LLC, Commercial and Government Entity (CAGE) 75Q65, for National Stock Number (NSN) 2520-01-592-8298, AXLE ASSEMBLY, AUTOMOTIVE, DRIVING, using Army Working Capital Funds (AWCF).

3. Description of Supplies/Services: The NSN 2520015928298, part number 3815350, CAGE 75Q65, AXLE ASSEMBLY, AUTOMOTIVE, DRIVING, is used on the Heavy Equipment Transporter (HET) M1070 A1. The deliverable for this contract is for a base quantity of 10 each, with an option quantity of 10 each, making a total quantity of 20 each. The total maximum value of this contract is [REDACTED]

Acquisition Method Code (AMC):

3 - Acquire, for the second or subsequent time, directly from the actual manufacturer.

Acquisition Method Suffix Code (AMSC):

R - The Government does not own the data or the rights to the data needed to purchase this part from additional sources. It has been determined to be uneconomical to buy the data or rights to the data. It is uneconomical to reverse engineer the part. This code is used when the Government did not initially purchase the data and/or rights. If only one source has the rights or data to manufacture this item, AMCs 3, 4, or 5 are valid. If two or more sources have the rights or data to manufacture this item, AMCs 1 or 2 are valid.

4. Authority Cited: The statutory authority permitting other than full and open competition for this effort is 10 United States Code (U.S.C.) 3204(a)(1) as implemented by the Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. Reason for Authority Cited:

This axle is found on the HET A1 platform. Axles are a critical component of the vehicle's drive train assembly, which transfers power generated by the engine from the transmission to the wheels. The wheels are mounted to the axle assembly, providing mobility to the vehicle. The HET A1 is designed and produced by Oshkosh. Oshkosh developed the system level requirements for the drive train system and possesses details regarding the design intent of this component. The performance characteristics, mounting details, materials, components, specifications, and certifications of this component are unknown. Oshkosh is the only vendor who possesses the technical data and [REDACTED]

expertise to manufacture this axle. The Government does not own the necessary intellectual property data rights to the pertinent technical data to enable competitive re-procurement. The axle is manufactured solely by Oshkosh, who owns all the data rights to the axle assembly. Only Oshkosh has the specific technical expertise and proper facilities to manufacture the axle. Additionally, Oshkosh has the history of building axle assemblies that allow the HET A1 to perform its mission. Oshkosh has manufactured axle assemblies that have successfully operated in the field and have met all mobility requirements outlined in the Army Technical Purchase Description (ATPD), such as being able to handle a 28,700lb load. Unlike other, commercially produced axles, this axle has been designed specifically for use on the HET A1. Oshkosh and the Government conducted developmental and operational testing during the vehicle's design and initial production phases, which proved that Oshkosh is capable of manufacturing an axle that meets the Government's performance and reliability, availability, and maintainability (RAM) requirements. If the Government were to purchase axle assemblies that have not undergone proper verification, mission failures and potential loss of mission critical functions to the vehicle could occur. Without this proper verification, it is likely that the axle assembly would not fit or function properly, which could result in failure, jeopardizing mission capability and fleet mobility.

Due to the lack of technical data available for the axle, reverse engineering is the only recourse for the Government to increase competition through development of a competitive build-to-print TDP. The Ground Vehicle Systems Center (GVSC) Heavy Tactical Vehicle Sustainment Team has estimated the effort that would be required to reverse engineer this axle based on conversations with Subject Matter Experts (SMEs) in Reverse Engineering, Test & Evaluation, and Quality Engineering. The reverse engineering process consists of procuring samples of the current part, tearing down the samples to understand and characterize the assembly's components, dimensions, and materials, and designing the reverse engineered part. Several prototypes would need to be fabricated. The process to reverse engineer the part and fabricate prototypes would cost an estimated minimum of \$1,000,000 with a lead time of 12 months, of which \$[REDACTED] would be used to procure samples, \$[REDACTED] would be used to characterize the part and develop the new design, and \$[REDACTED] would be used to fabricate prototypes. In addition, the prototype would have to undergo design validation and vehicle integration testing, which would cost an additional estimated minimum of \$[REDACTED] with an additional lead time of 24 months. The prototype testing would consist of a form, fit, and function test, a performance test, RAM testing, accelerated life testing, cold and hot environment testing, salt spray testing, and maximum load operation testing. To procure this axle competitively, a complete TDP would need to be developed, which would cost an estimated minimum of \$[REDACTED] with an additional lead time of 6 months. In total, reverse engineering, design validation, performance testing, and developing the TDP would require at least 42 months and an estimated minimum of \$3,500,000. To neglect design validation and integration testing would be a high risk as premature failures of this item in the field could jeopardize the Army's mission in a potentially adverse operational environment. Ultimately, there is a vital need for correct approved components to be contractually obtained under this procurement to maintain readiness and avoid vehicle dead-lining due to supply problems.

The Government would not be able to recover, through subsequent competition, the duplication of costs involved in reverse engineering the item to create a new and competitive TDP within a reasonable amount of time. At the current average usage rate of [REDACTED] units each year, a current unit cost of [REDACTED], and assuming [REDACTED] % cost savings achieved through competition, the in annual savings of \$ [REDACTED] would take over [REDACTED] years to offset the estimated \$3.5 million cost of reverse engineering the item and developing a competitive TDP.

6. Efforts to Obtain Competition:

- a. Effective competition: Due to the reasons cited in section 5 above the government is not pursuing competition for this item.
- b. Subcontracting competition: Unless the contract is awarded to a small business, the solicitation and resulting contract will include FAR clauses 52.219-8, "Utilization of Small Business Concerns"; 52.244-5, "Competition in Subcontracting"; 52.219-16, "Liquidated Damages—Subcontracting Plan"; and Defense Federal Acquisition Regulation Supplement (DFARS) 252.219-7003, "Small Business Subcontracting Plan (DoD Contracts)," to maximize small business opportunities and utilization.

7. Actions to Increase Competition: There are no plans to increase competition for future procurements due to the reasons cited above.

8. Market Research: A Sources Sought Notice was posted to SAM.gov on 05 October 2023, and was open for 15 days. No responses received.

Market research was conducted by the GVSC Heavy Tactical Vehicle Sustainment Team during October of 2023 to find suitable competition for the Oshkosh Axle Assembly. Axles are identified by characteristics such as weight rating and wheel end hardware, as well as interfaces with suspension and wheel/tire assemblies. The materials, dimensions, mounting details, and performance requirements are not available for this assembly. Due to the lack of a TDP or in-depth technical specifications of the sole-source part, the validity of researched axles is difficult to evaluate. Other manufacturers offer similar axles, but the form, fit, and/or function of alternatives are not identical to the sole-source part.

Keywords:

Heavy Truck Axle

HET Axle

Heavy Duty Axle

Websites:

<https://hdtransaxle.com/product/axletech-off-highway/planetary-steer-axles/>
<https://www.meritor.com/>
<https://www.dexteraxle.com/Products/Heavy-Duty>

Findings:

The results of this search did not produce any new companies that would provide a direct replacement of the dressed axle assembly that is 100% new and could be installed in the HET A1 without considerable time and effort to adapt the new components. Additionally, internet searches did not yield any products that meet the exact performance specs or would come installed with the proprietary fittings, parts, or weldments that Oshkosh specifically dresses the axles within order to fit to the HET A1 chassis.

9. Interested Sources: To date no additional sources have expressed interest in this item.

10. Other Facts: Procurement History:

Contract Number: SPRDL122C0019

Award Date: April 2022

Contractor: Oshkosh Defense LLC

Contract Value: \$399,599.76

Competitive Status: - FAR 6.302-1 only one responsible source

Contract Number: SPRDL117C0305//0001

Award Date: September 2017

Contractor: Oshkosh Defense LLC

Contract Value: \$511,202.88

Competitive Status: - FAR 6.302-1 only one responsible source

11. Technical Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [REDACTED]

Title: Branch Chief

Email: [REDACTED]@army.mil

Signature:

4/29/2024

X [REDACTED]
Signed by: [REDACTED]

12. Requirement Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my

knowledge and belief.

Name: [REDACTED]
Title: Branch Chief
Email: [REDACTED]@army.mil
Signature:

4/26/2024

X [REDACTED]

Signed by: USA

13. Fair and Reasonable Cost Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. This determination will be based on the results of a detailed cost and/ or price analysis. The contractor will be required to submit other than certified cost and pricing data as the expected value of this procurement falls under the TINA threshold. This item is commercial.

Name: Adam Reinbolt
Title: Contracting Officer
Email: adam.reinbolt@dla.mil
Signature:

4/29/2024

X Adam Reinbolt

Signed by: DLA

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Name: Adam Reinbolt
Title: Contracting Officer
Email: adam.reinbolt@gmail.com
Signature:

4/29/2024

X Adam Reinbolt

Signed by: DLA